



Assignment Report 3

DAMO-611-16 - Case Study 3 - Spring 2026 - Dr. Hany Osman - June 14, 2026

Group 3

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Ariana Lima

Joice Tolentino

Kelly Flor

Margi Patel

Table of Contents

1. Statement of Purpose.....	4
2. Project Scope.....	7
3. Design and Data Collection Methods.....	10
1. EIA – Global Brent Prices (Daily): The raw global oil dataset contained 3 fields and 10,333 rows.....	10
2. CEPEA – Brazil Ethanol Prices (Daily): The wholesale ethanol dataset initially comprised 4,043 rows across 3 fields.....	11
3. FRED – BRL/USD Exchange Rate (Daily): The exchange rate dataset comprised 4,239 rows across 2 fields.....	11
4. Exploratory Data Analysis.....	15
4.1 Descriptive Analysis.....	15
4.2 Brazilian Ethanol Prices, Brent Crude Oil, and BRL/USD Exchange Rate (2010–2026)	16
4.3 Correlation Heatmap: Ethanol Price, Brent Crude Oil, and BRL/USD Exchange Rate..	17
4.4 Monthly Average Ethanol Prices by Harvest Season (2010–2026).....	18
4.5 Brent Crude vs. Ethanol Price: Peacetime vs. 2025–2026 Geopolitical Conflict Period..	19
4.6 Temporal Rolling Correlation: Brent Crude Oil and Brazilian Ethanol Prices (2010–2026).	20
4.7 Synthesis.....	21
5. Methodology and Strategies.....	22
5.1 Data Preparation & Specification.....	22
5.2 Incremental ARIMAX model building.....	23
5.3 Testing the geopolitical shock effect.....	24
5.4 Model interpretation strategy.....	25
6. Model Building & Evaluation.....	27
6.1 Core ARIMAX parameters.....	27
6.2 Model 1: Baseline ARIMAX.....	28
6.3 Model 2: ARIMAX with Brent oil.....	28
6.4 Model 3: ARIMAX with Exchange Rate.....	30
6.5 Model 4: ARIMAX with Seasonal Factors.....	31
6.6 Model 5: ARIMAX with 2026 Geopolitical Shock.....	32
6.7 Structural Break Scenario Simulation: Additive vs. Multiplicative Shock.....	34
6.8 Isolated Wartime Forecast.....	35
6.9 Grid Search and Time-Series Cross-Validation (CV).....	37
6.10 Model Comparison Summary.....	38
6.11 Robustness and Sensitivity.....	40
6.11.1 Regime stability analysis.....	40
6.11.2 Temporal Robustness Check (2-week and 4-week Ahead Forecasting).....	41

6.12 Residual Diagnostic Metrics - Model 2 Brent.....	42
6.13 Final Forecast Projection.....	45
7. Limitations.....	47
8. Business Impact.....	50
8.1 Supply chain optimization: Petrobras and regional distributors.....	50
8.2 Retail pricing strategy: gas station operators.....	51
8.3 Inflation forecasting: Central Bank of Brazil.....	52
8.4 Subsidy Budget Planning: Ministry of Finance.....	53
8.5 Stakeholder Decision Framework - Summary.....	53
9. Conclusion.....	56
10. References.....	59

1. Statement of Purpose

The ongoing conflict between Iran and the United States, initiated in February 2026 with a joint US - Israel military operation that resulted in the closure of the Strait of Hormuz, has caused the largest oil supply shock on record, disrupting approximately 20% of global crude oil flows¹ and triggering volatility across international markets.

Brent crude prices recently increased more than 50% relative to the start of the year, and analysts forecast that the benchmark could average as high as USD 115 per barrel² whether critical oil and gas facilities suffer further damage. For emerging economies, such shocks transmit through complex channels, affecting domestic fuel prices, inflation, household welfare, and government fiscal policy.

Brazil presents an analytically distinctive case for studying these price transmission mechanisms. As the world's second-largest sugarcane ethanol producer³ and operator of the largest flex-fuel vehicle fleet^{4 5} globally, the country has developed a structural buffer against oil price shocks that most economies seem not to possess at scale. Tens of millions of Brazilian drivers can choose between filling their tank with 100% sugarcane-based ethanol or a gasoline blend that contains 30% of biofuel, a program launched in 1975 that has successfully evolved to reduce dependency on foreign oil.

Flex-fuel vehicles reached a record 76.2% share of Brazil's light vehicle fleet in circulation by 2023⁶, allowing consumers to choose between gasoline and hydrous ethanol (E100) based on relative prices. The practical consequence has been visible during the 2026 price shocks: while fuel prices rose between 30% and 40% in the United States, Brazilian prices increased by only 10% to 20%⁷, a difference attributed in part to the availability of ethanol and biodiesel as credible substitutes and to Petrobras's role in absorbing part of the cost pass-through.

¹ Dunn, C., & Barden, J. (2025, June 16).

² Dunn, C., & Barden, J. (2025, June 16).

³ Hayashi, T. (2024, August 31).

⁴ Consisting of vehicles capable of running on any combination of ethanol and gasoline.

⁵ Savarese, M. (2026, March 31).

⁶ Rodrigues, A. (2024, April 24).

⁷ The Economist. (2026, March 26).



The academic literature has confirmed the existence and complexity of the price transmission linkages at stake. Carpio (2019), using a vector error correction model with exogenous variables (VECX), demonstrated that there are significant long-term effects of oil prices on ethanol, gasoline, and sugar price forecasts in Brazil, and that in the short term both ethanol and gasoline price forecasts are more sensitive to oil price changes than sugar prices are, underscoring the centrality of the oil–ethanol relationship for consumer welfare analysis.

Atílio, Faria, Rodrigues, and Silva (2025), employing VAR and GVAR models across Brazil, China, India, and the United States from 2002 to 2023, found that oil supply shocks reduce both prices and ethanol production while demand shocks increase them, and confirmed that Brazilian and Indian ethanol markets are notably sensitive to international oil market dynamics.

Roland and Geman (2026) further established that spillover effects between crude oil, biofuels, and agricultural commodities are dynamic and asymmetric, intensifying precisely during periods of geopolitical tensions and energy policy uncertainty, notably the conditions that characterize the present analytical context.

It is important to note, however, that Brazil's insulation mechanism is structurally incomplete. The ethanol buffer operates exclusively at the gasoline substitution level: because flex-fuel vehicles cannot substitute ethanol for diesel, logistics, freight, and agricultural transport, costs remain fully exposed to international crude oil price volatility. Approximately 25% of Brazil's diesel consumption depends on imports⁸, making the country vulnerable to international price spikes along that channel.

Research shows that oil price increases first affect diesel costs, which then transmit to food and service prices with a lag, creating inflationary pressures that disproportionately affect lower-income households who spend the largest share of their income on food and transport. The World Bank's April 2026 Commodity Markets Outlook⁹ warned that, under a persistent disruption scenario, inflation in developing economies could rise to

⁸ Savarese, M. (2026, March 31).

⁹ World Bank. (2026, April 28).



5.8% in 2026 (a level exceeded only in 2022 over the past decade) with the poorest people being the most impacted.

In this context, the objectives of this project are therefore:

1. To develop a weekly ethanol price forecasting framework using an incremental ARIMAX model, in which Brent crude oil prices, the BRL/USD exchange rate, harvest seasonality, and geopolitical shock variables are added separately to assess the individual contribution to forecast performance.
2. To evaluate whether each exogenous variable provides significant predictive value, comparing model performance across specifications using out-of-sample RMSE, AIC, baseline comparison, and Diebold-Mariano tests.
3. To translate the weekly ethanol price forecasts into practical decision-making tools for key stakeholders, such as Petrobras, gas station retailers, the Central Bank, and the Ministry of Finance. The forecast can support weekly ethanol delivery planning and inventory allocation as well as help guide pump pricing, purchase timing, and promotional decisions. For the Central Bank and the Ministry of Finance, weekly forecasts can be aggregated into monthly indicators to support inflation monitoring and help distinguish temporary price spikes from more persistent structural increases, supporting better subsidy and fiscal planning during periods of oil market volatility.

Together, these objectives expect to combine time series forecasting, statistical testing, and economic interpretation to bridge the gap between descriptive ethanol price analysis and actionable energy policy in a globally important biofuel economy.

2. Project Scope

Building on the analytical motivations presented in the previous section, this project forecasts Brazilian ethanol prices and identifies which variable better contributes with the weekly prices variation in the Paulínia wholesale ethanol market. The project adopts a time-series forecasting approach that evaluates how ethanol prices interact with oil prices, seasonal supply demand, exchange rates, and how geopolitical conflict affects domestic ethanol prices dynamics.

The scope is limited to the weekly forecasting of Brazilian ethanol prices, using Paulinia wholesale ethanol price as the target variable. The main exploratory variables are: Brent crude oil price, BRL/USD exchange rate, harvest seasonality, and geopolitical conflict indicator. All variables will be aggregated to weekly frequency to ensure consistency throughout the modelling dataset.

Research Question: Which factors - Brent crude oil prices, the BRL/USD exchange rate, sugarcane harvest seasonality, and geopolitical shocks - contribute most to explaining the weekly variation in Brazilian ethanol prices?

Hypotheses:

H1o: Brent crude oil price **is not** a dominant predictor of Brazilian ethanol prices.

H1a: Brent crude oil price is a dominant predictor of Brazilian ethanol prices.

The coefficient of lagged Brent crude oil price will be examined to assess the direction of oil price transmission into ethanol prices.

H2o: The BRL/USD exchange rate **is not** a significant mediator of how oil shocks translate into domestic ethanol prices.

H2a: The BRL/USD exchange rate is a significant mediator of how oil shocks translate into domestic ethanol prices.

The coefficient of lagged BRL/USD exchange rate will be examined to assess the direction of currency transmission.

H3o: Sugarcane harvest seasonality **does not** explain additional variance beyond macroeconomic factors.

H3a: Sugarcane harvest seasonality explains additional variance beyond macroeconomic factors.

A sugarcane harvest season dummy coefficient will be examined to determine whether ethanol prices are systematically lower or higher during this season.

H4o: The 2026 geopolitical shock **does not** represent a structural break where the relationship between oil and ethanol intensifies.

H4a: The 2026 geopolitical shock represents a structural break where the relationship between oil and ethanol intensifies.

To evaluate the geopolitical shock, a conflict dummy coefficient will test if ethanol prices were systematically higher during the conflict period after controlling the previous variables.

The hypotheses are evaluated through an incremental ARIMAX modeling framework. Each successive model introduces a new explanatory variable while retaining the control variables and statistically relevant predictors identified in previous stages. Model performance is assessed primarily through out-of-sample Root Mean Squared Error (RMSE), supported by Akaike Information Criterion (AIC), Diebold-Mariano forecast comparison tests, expanding-window Time Series Cross-Validation, and an independent holdout test set. Hypotheses are evaluated based on both forecasting performance and



economic interpretation. Support is assessed through changes in out-of-sample RMSE, in-sample explanatory power (AIC), statistical testing, and the consistency of results with the proposed economic mechanisms.

Rationale: The research question establishes the foundation of this project by examining if Brazilian ethanol prices are driven mainly by their own historical behaviour and how external market conditions provide predictive information. By adding these variables through the ARIMAX framework, the project can evaluate how each factor improves forecast accuracy compared with a baseline ARIMAX model and a naive benchmark, allowing the analysis to move towards a more practical forecasting structure, supporting decision-making for fuel supply planning, and policy response.

3. Design and Data Collection Methods

The data for this study was collected from three different sources. Table 3.1 describes the key selected variables, exposing each variable type, description and source.

Table 3.1 - Collected Variables

Variable Original Name	Variable Final Name	Type	Operational Definition	Source
Europe Brent Spot Price FOB (Dollars per Barrel)	BRENT PRICE USD/L	Independent (continuous)	Global crude oil price (US\$/barrel)	EIA ¹⁰
À Vista R\$	ETHANOL BRL/L	Dependent (continuous)	Hydrous ethanol price - Paulínia (R\$/liter or US\$/liter)	CEPEA ¹¹
À Vista US\$	ETHANOL USD/L			
RXI_N.B.BZ	EXCHANGE RATE	Independent (continuous)	Brazilian Real to US Dollars exchange rate (BRL/USD)	FRED ¹²

All selected variables have a daily frequency, and the analysis spans from February 1, 2010 to April 30, 2026. This specific timeframe was dictated by the historical availability of the ethanol prices; although the raw data begins on January 25, 2010, the dataset was truncated aiming at including only complete calendar months to ensure chronological consistency.

The software Tableau Prep Builder was used to perform the data cleaning and standardization. The applied steps for each source are detailed below:

1. EIA – Global Brent Prices (Daily): The raw global oil dataset contained 3 fields and 10,333 rows.

- **Dimensionality Reduction:** Following initial header adjustments, one irrelevant field was removed to streamline the data structure (resulting in 2

¹⁰ EIA (U.S. Energy Information Administration) - Available at https://www.eia.gov/dnav/pet/PET_PRI_SPT_S1_D.htm

¹¹ CEPEA (Centro de Estudos Avançados em Economia Aplicada) - Available in Portuguese at <https://www.cepea.org.br/br/indicador/etanol-diario-paulinia.aspx>

¹² FRED (Federal Reserve Bank of St. Louis) - Available at <https://www.federalreserve.gov/releases/h10/current/>

fields). The timeline was subsequently filtered to the standard February 2010 – April 2026 window, reducing the dataset to 4,153 rows with no duplicate entries. In addition, a price conversion was applied from barrel to liter to ensure a common unit of measure across the datasets.

- **Missing Data Imputation:** The "BRENT PRICE" variable exhibited a missingness rate of <1%. This occurs because the EIA republishes spot prices from Refinitiv; consequently, days characterized by thin trading or international market closures generate missing values in the FRED system. To resolve this without introducing look-ahead bias, a Last Observation Carried Forward (LOCF) methodology was executed utilizing a calculated field.

2. CEPEA – Brazil Ethanol Prices (Daily): The wholesale ethanol dataset initially comprised 4,043 rows across 3 fields.

- **Standardization:** Preparatory steps included the translation and adjustment of headers from Portuguese to English, alongside mandatory data type conversions, as well as price conversion from m³ to liter.
- **Filtering & Validation:** The dataset was filtered strictly to the primary analytical timeframe of February 1, 2010, to April 30, 2026. This yielded a final count of 4,029 rows. The data was verified to contain zero duplicate entries and no missing values.

3. FRED – BRL/USD Exchange Rate (Daily): The exchange rate dataset comprised 4,239 rows across 2 fields.

- **Standardization:** Headers were adjusted for consistency, and appropriate data type conversions were applied. The dataset was checked and confirmed to contain no duplicates.
- **Missing Data Imputation:** The "EXCHANGE RATE" field presented a 4% missing value rate, primarily corresponding to global banking holidays and non-trading days. In accordance with the treatment of the macroeconomic oil dataset, these temporal gaps were resolved by applying the LOCF technique via a calculated field.



Finally, with the goal of forecasting weekly ethanol prices, the datasets from all three sources were aligned to a weekly frequency starting on Sundays. The variables were aggregated using weekly averages, ensuring the same temporal unit and preventing inconsistencies. With this, the three individual datasets culminated into the final 848 rows each that were merged using an inner join with the date variable - now the date that starts each week (START DATE) - as a common key.

The outliers analysis using boxplots - Figure 3.1 - revealed that while the standalone domestic ethanol price (BRL) and the BRL/USD exchange rate exhibited normal historical distributions with zero mathematical outliers, the USD-ethanol price contained 29 outliers. They were found to come from the following conditions:

- **The 2011 Crisis (High Outliers):** Between February and October 2011, a period referred to as “global currency war”¹³, the Brazilian Real was incredibly strong (trading around 1.5 to 1.7 BRL per USD). Because the denominator was so small, converting standard domestic ethanol prices into USD made it look like ethanol was astronomically expensive on the global stage.
- **The Early 2020 COVID Crash (Low Outliers):** Between March and May 2020, two things happened at once: domestic ethanol prices dropped due to lockdowns, and the Brazilian Real completely crashed (spiking to 5.8 BRL per USD). Dividing a shrinking domestic price by a massively inflated exchange rate created extreme low-end outliers.
- **The 2022 Inflation Spike (High Outliers):** In April 2022, domestic ethanol prices spiked to around 0.84 USD/L due to post-pandemic inflation¹⁴ and supply chain issues, instantly triggering upper-bound outliers.

As all the 29 outliers were likely due to real-world events, they couldn't be just deleted. Instead, Intervention Analysis was applied, with the creation of “dummy” variables that specifically flagged the time period of the three historical events. This way, including these dummy variables into the model mathematically instructs it to calculate a separate

¹³ Available at <<https://g20.utoronto.ca/analysis/111109-cannesperf-brazil.html>>

¹⁴ Available at <<https://www.macrotrends.net/global-metrics/countries/bra/brazil/inflation-rate-cpi>>

coefficient for those historical crises. This neutralizes the financial noise, allowing the model to learn the true, "normal" relationship between global Brent Crude and Brazilian ethanol while also maintaining the chronological chain of the data for the time-series forecasting.

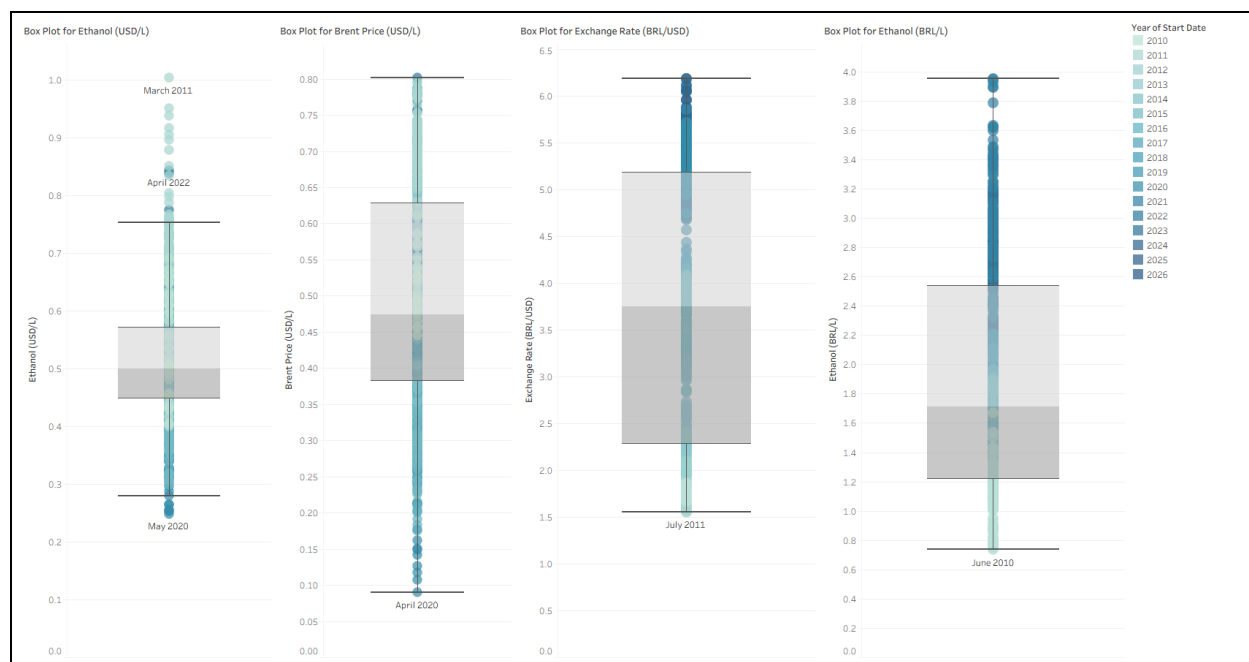


Figure 3.1 - Data Distribution

Alongside with the new variables from Intervention Analysis, it was noticed the need of more data engineering for the model development, therefore described in Table 3.2.

Table 3.2 - Engineered Variables

Variable Name	Type	Operational Definition	Source
DUMMY 2011 CRISIS	Independent (binary)	Intervention Analysis for outliers; Equals 1 during the “currency war” and 0 otherwise.	Constructed based on BRL’s overevaluation in 2011 ¹⁵
DUMMY 2020 COVID	Independent (binary)	Intervention Analysis for outliers; Equals 1 during the pandemics crash and 0 otherwise.	Constructed based on early COVID Crash in Brazil ¹⁶
DUMMY 2022 INFLATION	Independent (binary)	Intervention Analysis for outliers; Equals 1 during 2022 inflation period and 0 otherwise.	Constructed based on Brazil’s inflation ¹⁷
BRENT LAG1	Independent (continuous)	Brent crude oil price lagged by one week to test delayed transmission into ethanol prices.	Constructed from EIA data
EXCHANGE RATE LAG1	Independent (continuous)	BRL/USD exchange rate lagged by one week to test delayed transmission into ethanol prices.	Constructed from FRED data
HARVEST DUMMY	Independent (binary)	Equals 1 during Brazil’s sugarcane harvest season and 0 otherwise.	Constructed from the CONAB calendar ¹⁸
CONFLICT DUMMY	Independent (binary)	Equals 1 during the US and Iran geopolitical shock period and 0 otherwise.	Constructed variable from current news ¹⁹

¹⁵ Available at <<https://www.elibrary.imf.org/display/book/9781484364987/ch005.xml>> and <<https://g20.utoronto.ca/analysis/111109-cannesperf-brazil.html>>

¹⁶ Available at <<https://repositorio.cepal.org/server/api/core/bitstreams/e846340c-7f8a-4f05-8b66-8f63915aecfa/content>>

¹⁷ Available at <<https://www.macrotrends.net/global-metrics/countries/bra/brazil/inflation-rate-cpi>>

¹⁸ Lisbinski, F. C. (2025)

¹⁹ Available at <<https://www.nytimes.com/live/2026/05/29/world/iran-war-us-trump-deal>>

4. Exploratory Data Analysis

Python was used for descriptive analysis and Tableau Visualizations were used to support the variable selection process before model estimation. Rather than serving only as descriptive charts, these visuals helped identify whether Brent crude oil, the BRL/USD exchange rate, harvest seasonality, and the geopolitical conflict period had enough theoretical and visual relevance to be tested in the ARIMAX framework. The charts provide an exploratory foundation for understanding why each variable was included in the analysis.

4.1 Descriptive Analysis

	EXCHANGE_RATE	BRENT_PRICE_USD/L	ETHANOL_USD/L	ETHANOL_BRL/L
count	848.000000	848.000000	848.000000	848.000000
mean	3.740670	0.489514	0.514922	1.874774
std	1.408603	0.152499	0.107098	0.737365
min	1.551040	0.089542	0.248584	0.737300
25%	2.280860	0.381946	0.448843	1.216600
50%	3.751860	0.474088	0.500758	1.711550
75%	5.173115	0.628231	0.570812	2.531925
max	6.183740	0.801311	1.003612	3.952875

Figure 4.1.1 - Summary Statistics

The provided descriptive statistics - Figure 4.1.1 - summarizes the 848 weekly observations, capturing the interplay between the Brazilian ethanol market and global energy benchmarks. The data reveals a significant spread in currency and commodity valuations, with the EXCHANGE RATE exhibiting substantial volatility (mean of 3.74, ranging from 1.55 to 6.18), which directly impacts the cost parity of imported inputs. Notably, the price of ethanol, whether denominated in USD (mean 0.51) or BRL (mean 1.87), shows a high degree of variance - evidenced by the substantial standard deviation in the ETHANOL BRL/L variable - reflecting the market's sensitivity to both the underlying Brent crude oil price and domestic currency fluctuations. This statistical

profile confirms that the ethanol market operates under heterogeneous regimes, where the interplay between the exchange rate and global energy benchmarks necessitates the use of exogenous regressors and structural break controls to accurately model price behavior over time.

4.2 Brazilian Ethanol Prices, Brent Crude Oil, and BRL/USD Exchange Rate (2010–2026)

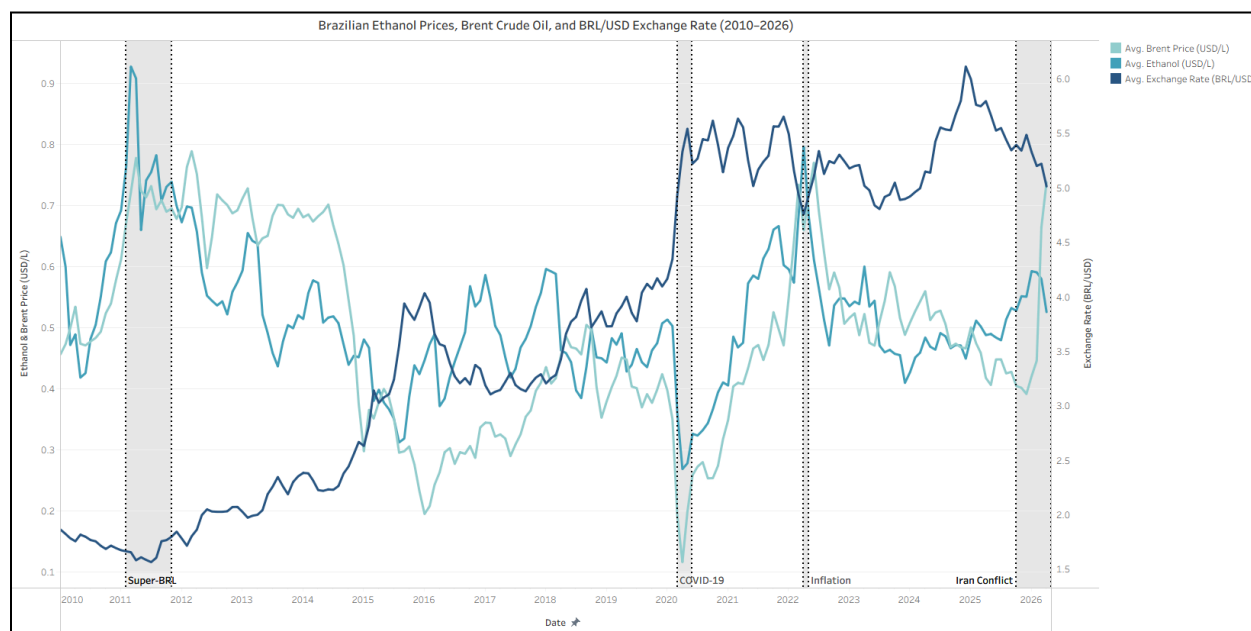


Figure 4.2.1 - Brazilian Ethanol Prices, Brent Crude Oil, and BRL/USD Exchange Rate (2010–2026)

The chart provides the broad historical context of the project by plotting ethanol prices, Brent crude oil prices, and the BRL/USD exchange rate over time. This chart helps show how the main variables behaved from 2010 to 2026 and highlights major market events such as the Super-BRL period, COVID-19, inflationary pressure, and the Iran conflict period.

It shows that ethanol prices do not move in isolation. Across the historical period, ethanol prices appear to respond to both domestic and international market conditions. Brent crude oil captures the global energy price channel, while the exchange rate

captures the currency channel through which dollar-denominated shocks may affect Brazil's domestic market.

For the research question, this chart supports the decision to include both Brent crude oil and the exchange rate as candidate exogenous variables in the ARIMAX model. It also introduces the idea that some periods, especially crisis periods, may behave differently from normal market conditions.

4.3 Correlation Heatmap: Ethanol Price, Brent Crude Oil, and BRL/USD Exchange Rate

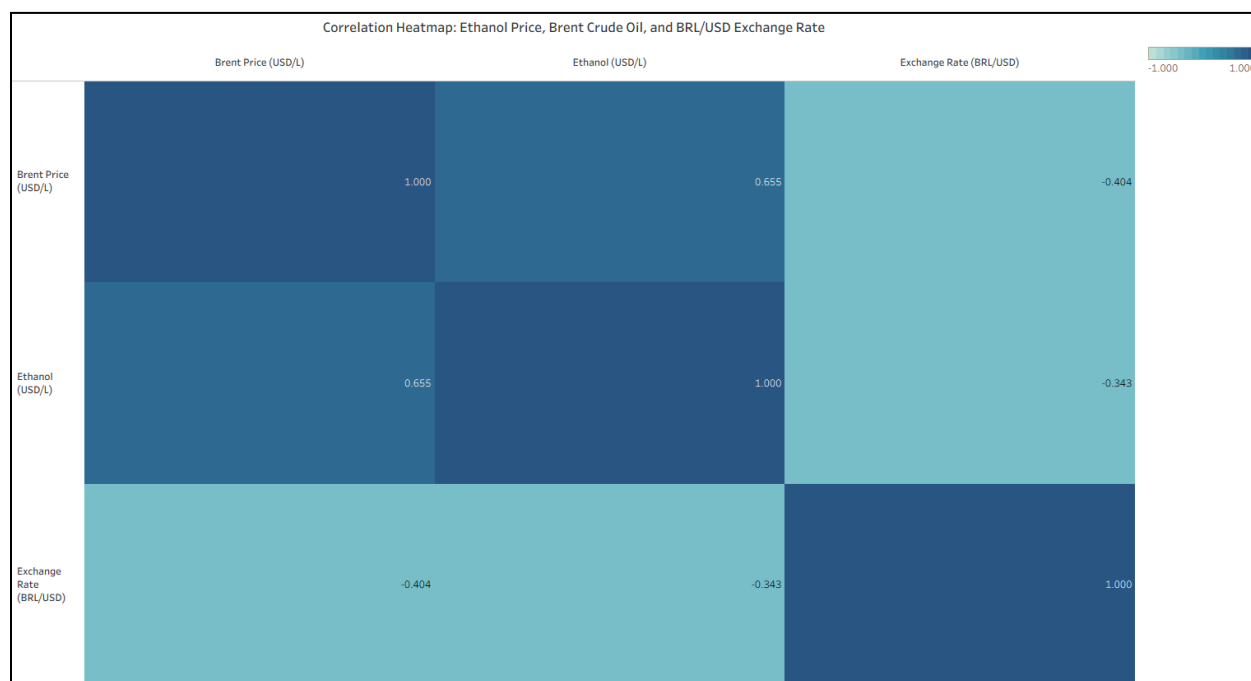


Figure 4.3.1 - Correlation Heatmap: Ethanol Price, Brent Crude Oil, and BRL/USD Exchange Rate

The correlation heatmap provides a first quantitative view of the relationship between the main variables. The chart shows a positive correlation between Brent crude oil prices and ethanol prices, supporting the expectation that international oil prices may provide useful information for explaining ethanol price movements.

The heatmap also shows the relationship between the exchange rate and the other variables. Since Brent is traded internationally in U.S. dollars, the BRL/USD exchange

rate is theoretically important for understanding how global price shocks are translated into the Brazilian market. Even though correlation does not prove causality, it helps identify which variables are worth testing in the forecasting model.

This visualization supports H1o as Brent shows a clear relationship with ethanol prices. It also supports the inclusion of H2o, because the exchange rate may act as a transmission mechanism between global oil markets and domestic fuel prices.

4.4 Monthly Average Ethanol Prices by Harvest Season (2010–2026)

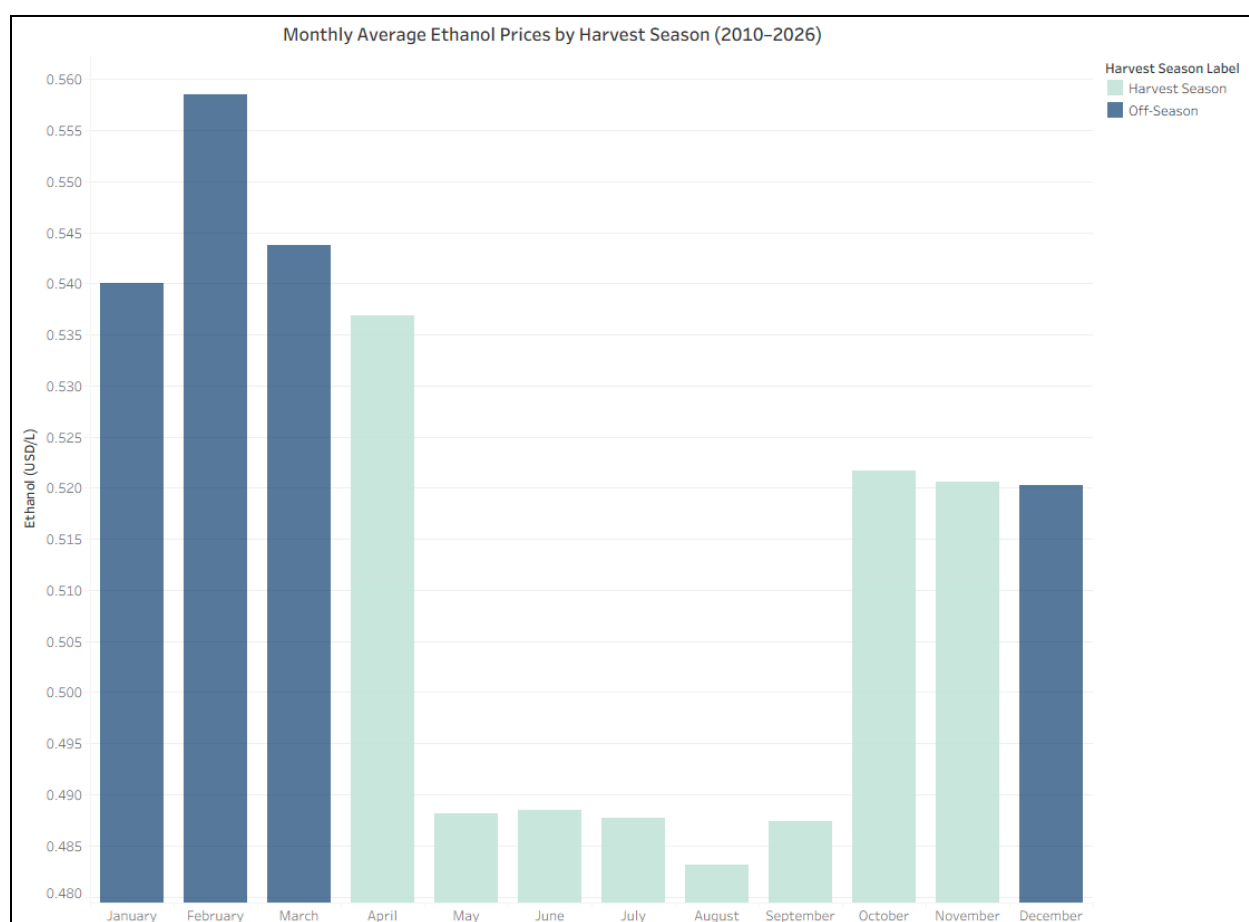


Figure 4.4.1 - Monthly Average Ethanol Prices by Harvest Season (2010–2026)

The seasonal subseries plot was used to examine if ethanol prices exhibit a recurring monthly pattern throughout the year. This is relevant as Brazilian ethanol production is closely connected to the sugarcane harvest cycle. If prices behave differently during

harvest and off-season months, then harvest seasonality may help explain week-to-week variation in ethanol prices.

Ethanol prices vary across months, suggesting seasonality in the data. It should be noted that the vertical axis of the chart does not start at zero; this adjustment was applied to amplify and improve the visibility of the data differences, which were not as distinct in a zero-indexed format. This provides a visual justification for testing the harvest dummy in the ARIMAX model. Even though the visualization alone does not confirm that harvest seasonality improves forecasting performance, it shows that the variable is theoretically and visually relevant enough to be tested.

The plot supports the motivation for H3o, proposing that harvest seasonality may explain additional price variation beyond macroeconomic factors.

4.5 Brent Crude vs. Ethanol Price: Peacetime vs. 2025–2026 Geopolitical Conflict Period

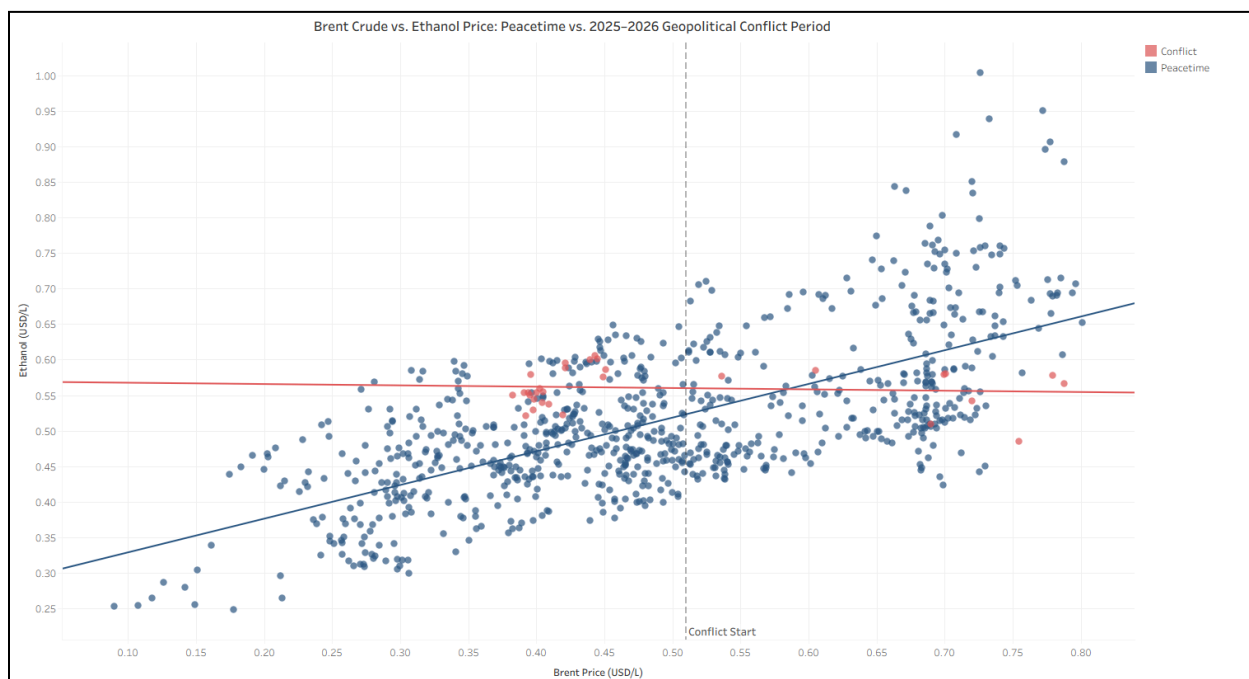


Figure 4.5.1 - Brent Crude vs. Ethanol Price: Peacetime vs. 2025–2026 Geopolitical Conflict Period

The regime-separated scatter plot compares the relationship between Brent crude oil prices and ethanol prices during normal market periods and the conflict period. This chart is important because it helps visualize whether the Brent–ethanol relationship changes under geopolitical stress.

During normal periods, the relationship between Brent and ethanol follows a more stable pattern. However, the conflict-period observations allow examining whether the market behaves differently when global oil risk increases. This is directly connected to the structural break hypothesis, because a geopolitical shock may not simply increase prices; it may also change the strength of the relationship between Brent and ethanol.

This chart supports H4o by providing a visual foundation for testing whether the 2025–2026 conflict period altered the oil-to-ethanol transmission mechanism and justifying why the model later compares additive and multiplicative conflict specifications.

4.6 Temporal Rolling Correlation: Brent Crude Oil and Brazilian Ethanol Prices (2010–2026)

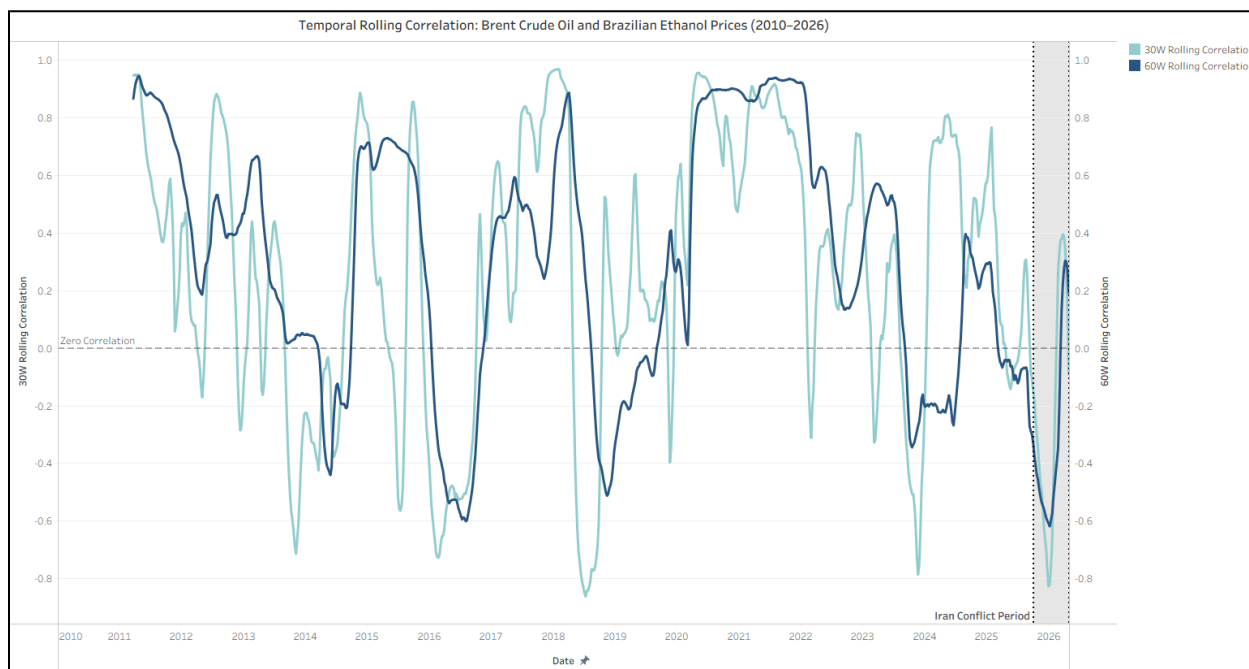


Figure 4.6.1 - Temporal Rolling Correlation: Brent Crude Oil and Brazilian Ethanol Prices (2010–2026)



A simple static relationship may not fully capture the behavior of the market, so the rolling correlation chart shows that the relationship between Brent crude oil and ethanol prices is not constant over time. Instead, the strength and direction of the relationship change over time.

Since the Brent–ethanol relationship changes over time, the model needs to account for dynamic patterns, lagged effects, and possible structural changes. To better support the analysis, a time-series forecasting framework was chosen rather than relying solely on simple correlation or static regression, reinforcing the idea that Brent crude oil is relevant, but its effect may depend on the market regime. It also supports the need to test the geopolitical shock hypothesis, since periods of stress may change how strongly oil prices transmit into ethanol prices.

4.7 Synthesis

The visualizations support the logic of the ARIMAX modeling framework. The time-series and correlation charts justify the inclusion of Brent crude oil and the exchange rate. The seasonal subseries plot explains why harvest seasonality was tested. The regime-separated scatter plot and rolling correlation chart support the need to examine whether the period of geopolitical conflict created a structural break in the oil-to-ethanol relationship.

Together, these visuals show that the selected variables were not chosen randomly. Each one has a theoretical and exploratory basis. The next section uses the ARIMAX model to test whether these visual patterns translate into measurable improvements in forecasting.

5. Methodology and Strategies

To capture the dynamics of the Brazilian ethanol market, this study employs an Autoregressive Integrated Moving Average with Exogenous Regressors (ARIMAX) framework. This model was selected for its capacity to integrate both the internal momentum of price series (autoregressive components) and the external influence of macroeconomic drivers. The analysis was conducted using Python and libraries like scikit-learn, statsmodels and scipy.

The methodological approach of this study is structured as a hierarchical analytical stack. Rather than treating the hypotheses as isolated inquiries, the framework is designed from sequentially deconstructing market noise, to isolating the structural mechanics of the 2026 geopolitical shock.

5.1 Data Preparation & Specification

To ensure the statistical validity of the ARIMAX framework, the time-series properties of the variables were examined through stationarity testing. The Augmented Dickey-Fuller (ADF) test was used to assess whether the series exhibited a stable mean and variance over time. In addition, Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were reviewed to provide an initial indication of the autoregressive (p) and moving-average (q) structures present in the data.

Based on these preliminary diagnostics, an initial ARIMAX specification was established. Subsequently, a systematic grid search combined with expanding-window Time Series Cross-Validation was conducted to evaluate alternative ARIMA parameter combinations and validate the robustness of the selected structure. This procedure allowed candidate models to be assessed under multiple training and validation windows while preserving the chronological order of observations and avoiding look-ahead bias.

The dataset was partitioned using a chronological train-test split, with 80% of historical observations reserved for model training and the remaining 20% dedicated to out-of-sample testing. This ensures that the temporal ordering of observations is

preserved, and that the model's performance is evaluated against unseen data, mitigating the risk of look-ahead bias.

To assess predictive accuracy, the model was benchmarked against two standards:

- **Information Criteria (AIC):** Used to penalize model complexity, ensuring the selection of the most parsimonious model architecture.
- **Error Metric (RMSE):** The Root Mean Squared Error was employed as the primary metric for quantifying the magnitude of forecast deviation during the testing interval.

The model order selection combined traditional time-series diagnostics with a systematic hyperparameter search procedure to identify statistically appropriate model structures while maintaining forecast robustness.

5.2 Incremental ARIMAX model building

The modeling strategy begins with a baseline autoregressive model, utilizing only historical ethanol price data. This baseline serves as the fundamental performance benchmark, against which the predictive utility of all subsequent exogenous variables is assessed.

To maintain internal consistency across the study, the baseline model incorporates the three indicator variables (dummy variables), previously presented in section 3, to control for historical structural breaks unrelated to the primary macroeconomic hypotheses:

- **DUMMY 2011 CRISIS (Feb–Oct 2011)**
- **DUMMY 2020 COVID (Mar–May 2020)**
- **DUMMY 2022 INFLATION (Apr 2022)**

These outlier controls remain constant across every subsequent iteration of the ARIMAX framework. This "fixed-control" approach ensures that any observed change in model performance can be attributed strictly to the incremental addition of new predictors. Consequently, rather than employing an "all-at-once" variable integration, we



utilize an incremental modular approach. By introducing exogenous variables (such as Brent Crude and exchange rates) one at a time, we are able to isolate and quantify the specific contribution of each regressor to the overall forecasting architecture.

The model sequence follows this order:

1. **ARIMAX baseline:** Ethanol price history only.
2. **ARIMAX with Brent (H1):** Integrates lagged Brent Crude oil prices, serving as the primary proxy for global energy commodity parity.
3. **ARIMAX with exchange rate (H2):** Incorporates the lagged BRL/USD exchange rate to evaluate how currency volatility and export parity influence domestic pricing. This model retains lagged Brent Crude only if statistically significant.
4. **ARIMAX with harvest seasonality (H3):** Introduces a harvest season dummy variable to account for the cyclical supply shocks inherent in the sugarcane production cycle. This model preserves significant exogenous predictors from the previous model.
5. **ARIMAX with geopolitical shock (H4):** The final specification, which introduces the structural break variables. This model evaluates the market's response to the 2026 Strait of Hormuz closure. Again, former significant predictors are also retained.

The Brent and exchange rate variables were lagged by one week because changes in global oil prices and currency markets may not transmit immediately into domestic ethanol prices. The harvest variable was designed to capture seasonal supply-side patterns, while the conflict dummy is expected to capture abnormal market conditions during the geopolitical shock period.

5.3 Testing the geopolitical shock effect

The evaluation of geopolitical risk presented a unique methodological challenge. Due to the chronological train-test split, the 2026 conflict period occurred entirely within the out-of-sample testing window. Consequently, the conflict indicator exhibited no variation

within the training dataset, preventing the ARIMAX framework from directly estimating its effect through conventional parameter learning.

Because the model could not learn a historical relationship between conflict conditions and ethanol prices, a scenario-based intervention analysis was adopted. This approach allowed the study to evaluate how a major geopolitical disruption might affect market behavior under alternative assumptions regarding price transmission mechanisms.

Two intervention frameworks were examined against the "peacetime" baseline:

1. **Additive Simulations:** Assume that geopolitical disruptions introduce a fixed price premium regardless of the underlying market level.
2. **Multiplicative Simulations:** Assume that geopolitical disruptions alter the sensitivity of ethanol prices to existing market drivers, resulting in proportional changes to forecasted values.

To evaluate these alternative structures, simulated intervention effects were applied to the baseline forecasting framework and compared against observed market behavior during the conflict period.

In addition to evaluating forecast performance across the full testing horizon, a localized wartime assessment was conducted. This supplementary analysis isolated the conflict period and calculated forecasting error exclusively during the weeks affected by the geopolitical disruption. By separating wartime observations from the broader testing sample, the study was able to assess which intervention framework most accurately represented market behavior under crisis conditions.

5.4 Model interpretation strategy

For each variable added to the model, the analysis examines whether it improves forecast accuracy and whether the estimated coefficient associated with that variable provides a meaningful economic interpretation. This includes the coefficients for lagged Brent crude oil, lagged BRL/USD exchange rate, the harvest season dummy, the additive conflict dummy, and the conflict-Brent interaction term.



Brent crude oil will be interpreted as the global energy price channel, the BRL/USD exchange rate as the currency transmission channel, and the harvest seasonality will represent domestic supply conditions. The geopolitical shock variable, in turn, will indicate whether abnormal global market stress shifts ethanol prices or intensifies the oil-to-ethanol transmission mechanism.

Because the primary objective of this project is forecasting, model assessment is centered on out-of-sample predictive performance. Forecast accuracy metrics are used to compare competing specifications against the baseline model, while information criteria provide supplementary evidence regarding model parsimony and in-sample fit. Time-series cross-validation and rolling-origin forecasting procedures are further employed to evaluate the stability and robustness of model performance across different forecasting horizons.

In addition to forecast accuracy, statistical comparison tests and residual diagnostics are used to assess model adequacy. These procedures help determine whether improvements in predictive performance are statistically meaningful and whether the model sufficiently captures the underlying temporal structure of the data.

Consequently, the preferred specification is not necessarily the most complex model, but rather the one that demonstrates the strongest combination of forecasting performance, robustness, statistical adequacy, and economic plausibility.

6. Model Building & Evaluation

6.1 Core ARIMAX parameters

The ADF test presented a statistic of -3.2505 and a p-value of 0.0172. As the p-value is below the 5% significance level, the null hypothesis of non-stationarity was rejected. The differencing parameter for the initial ARIMAX was then initially set to $d = 0$, indicating that the seminal ethanol price is stationary in level form.

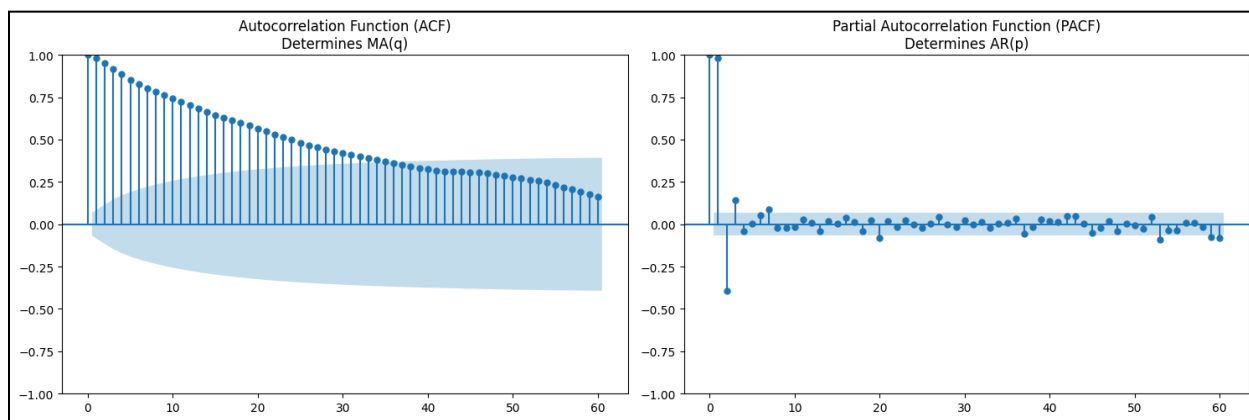


Figure 6.1 - ACF and PACF charts

The plots on Figure 6.1.1 were used to reveal how prices from past weeks correlate with current prices, allowing to determine how many historical steps (lags) the model needs to look back to make an accurate prediction. The ACF shows that there is no sharp moving average cutoff, just a gradual geometric decay, which sets the moving average parameter as zero ($q = 0$). As for the PACF, lag 4 is the first one to drop inside the blue confidence interval, indicating that the ethanol price from three weeks ago significantly impacts the price today, but the price from four weeks ago does not add any new predictive power. Therefore, $p = 3$. Based on these preliminary diagnostics, an ARIMAX(3,0,0) specification was adopted as the initial model architecture for the hypothesis-testing phase of the study.

To further validate the model structure, a systematic grid search was subsequently conducted across multiple (p,d,q) combinations using expanding-window time-series cross-validation. This procedure was used to evaluate alternative ARIMA specifications

and assess the robustness of the preliminary parameter selection obtained from the ACF and PACF analysis. The resulting optimized specification was later compared against the original ARIMAX(3,0,0) framework using out-of-sample forecasting performance and model evaluation metrics.

6.2 Model 1: Baseline ARIMAX

```
--- Baseline ARIMAX(3,0,0) Model ---  
Baseline AIC: -3425.2527  
Baseline Test RMSE: 0.0588
```

Figure 6.2.1 - Baseline ARIMAX Metrics

The baseline ARIMAX(3,0,0) model, fitted on the training sample with the three structural-break dummy variables as exogenous regressors, produced an AIC of -3425.2527 and a RMSE of 0.0588. This result becomes the baseline for the next models, as it shows how well ethanol prices can be forecast using only their own historical pattern and the control dummy variables. The RMSE of 0.0588 means that the baseline model already tracks weekly ethanol prices with a relatively small forecasting error. However, this model does not include external market signals yet, so the next step is to test whether adding variables such as Brent crude oil can reduce this error and improve out-of-sample predictive performance.

6.3 Model 2: ARIMAX with Brent oil

H1_A: Brent crude price is a dominant predictor of ethanol prices

In order to test whether global oil prices provide predictive information for Brazilian ethanol prices, the second model added lagged Brent crude oil prices, represented as *Brent_LAG1*, to the baseline ARIMAX specification.

```
--- Model 2: ARIMAX with Brent Crude ---  
ARIMAX (Brent) AIC: -3421.3766  
ARIMAX (Brent) Test RMSE: 0.0575  
  
AIC Improvement over Baseline: -3.8761  
RMSE Improvement over Baseline: 0.0013 (2.24%)  
  
--- Diebold-Mariano Test: Baseline vs. Model 2 ---  
DM Statistic: 4.6952  
p-value:      0.0000  
Conclusion: H1a Validated. Brent Crude significantly IMPROVES forecast accuracy.
```

Figure 6.3.1 - Model 2 Metrics

On one hand, adding lagged Brent crude oil to the model introduced a small in-sample penalty. The baseline model achieved a lower AIC of -3425.2527, compared with -3421.3766 for Model 2. This means that, when looking only at historical fit and model complexity, the simpler baseline model is still preferred, given that a lower (more negative) AIC is considered better for this metric. In other words, Brent adds some complexity that is not fully rewarded by the AIC during the training phase.

On the other hand, the out-of-sample results show that Brent improves the model's forecasting performance. Model 2 reduced the test RMSE from **0.0588** to **0.0575**, representing a 2.24% improvement over the baseline. Since the main objective of this project is prediction, this result is more important than the AIC penalty. The Diebold-Mariano test also supports this result, with a p-value < 0.05, indicating that the improvement in forecast accuracy is statistically significant.

Therefore, H1a is supported from a predictive perspective. Even though the baseline model is slightly better for summarizing historical data, the Brent-based model performs better when forecasting unseen data. This suggests that lagged Brent crude oil provides useful information for future ethanol price movements and can be treated as an external predictor in the final forecasting framework.

6.4 Model 3: ARIMAX with Exchange Rate

H2_A: The BRL/USD exchange rate is a significant mediator of how oil shocks translate into domestic prices.

In order to test whether exchange rate movements improve the model's ability to explain and forecast Brazilian ethanol prices beyond the information already provided by lagged Brent crude oil prices, the lagged BRL/USD exchange rate was added to Model 3. As Brent crude oil is priced internationally in U.S. dollars, the BRL/USD exchange rate is theoretically relevant because currency depreciation can amplify the domestic impact of global oil price shocks.

Model 3 yielded an AIC of -3442.2246, compared with -3421.3766 for the Brent-only model. This improvement of approximately 20.85 points suggests that the exchange rate adds meaningful explanatory power when fitting the historical data. However, the out-of-sample forecasting results were substantially weaker. The RMSE increased from 0.0575 in Model 2 to 0.1328 in Model 3, representing a deterioration of approximately 131.12% relative to the Brent-only model. This indicates that, although the exchange rate helps explain historical ethanol price movements, it does not improve forecast accuracy on unseen data. In fact, the higher RMSE suggests that the exchange rate may introduce instability when used for future prediction.

```
--- Model 3: ARIMAX with Brent + Exchange Rate ---  
ARIMAX (Brent+FX) AIC: -3442.2246  
ARIMAX (Brent+FX) Test RMSE: 0.1328  
  
AIC Improvement over Model 2: 20.8479  
RMSE Improvement over Model 2: -0.0753 (-131.12%)  
  
--- Diebold-Mariano Test: Model 2 vs. Model 3 ---  
DM Statistic: -16.7099  
p-value: 0.0000
```

Figure 6.4.1 - Model 3 Metrics

With a p-value < 0.05 , the Diebold-Mariano test indicates that the difference in forecast accuracy between Model 2 and Model 3 is statistically significant. And since Model 3 has a higher RMSE, this difference should be interpreted as evidence that the exchange rate model performs significantly differently, not that it performs better. Therefore, H2a is supported from an explanatory perspective. The BRL/USD exchange rate acts as a transmission channel through which dollar-denominated oil shocks affect Brazilian ethanol prices, providing meaningful economic interpretation and improving the model's explanatory power. However, the inclusion of the exchange rate increased out-of-sample RMSE, indicating that it does not improve forecasting performance. As a result, while the exchange rate helps explain historical price movements and the transmission of oil shocks into the domestic market, it is not retained as a useful forecasting variable in the final predictive model.

6.5 Model 4: ARIMAX with Seasonal Factors

H3_A: Harvest seasonality explains additional variance beyond macroeconomic factors.

The fourth model added a binary harvest season dummy to the Brent-only ARIMAX specification. This variable was included to test whether Brazil's sugarcane harvest season provides additional predictive information beyond lagged Brent crude oil prices. Since ethanol production is closely linked to sugarcane supply, the harvest season was expected to capture domestic supply-side effects that could place downward pressure on ethanol prices.

Model 4 produced an AIC of -3419.6672, compared with -3421.3766 for the Brent-only Model 2, suggesting the harvest dummy adds complexity without improving in-sample model efficiency.

The out-of-sample results also show a slight deterioration. The RMSE increased from 0.0575 in Model 2 to 0.0577 in Model 4, with a small decline in forecast accuracy of approximately 0.39%. Although this change is relatively small, it indicates that adding

the harvest-season dummy does not provide additional predictive value beyond the information already captured by the Brent-only model.

```

--- Model 4: ARIMAX with Brent + Harvest ---
ARIMAX (Brent+Harvest) AIC: -3419.6672
ARIMAX (Brent+Harvest) Test RMSE: 0.0577

AIC Improvement over Model 2: -1.7094
RMSE Improvement over Model 2: -0.0002 (-0.39%)

--- Diebold-Mariano Test: Model 2 vs. Model 4 ---
DM Statistic: -3.0537
p-value:      0.0023

```

Figure 6.5.1 - Model 4 Metrics

The p-value of 0.0023, from The Diebold-Mariano test, means that the difference in forecast errors between Model 2 and Model 4 is statistically significant. However, as Model 4 has a higher RMSE, this indicates that the harvest dummy worsens forecast performance rather than improves it.

These results do not support H3a. Although harvest seasonality is theoretically relevant for the Brazilian ethanol market, it does not explain additional variance beyond the macroeconomic factors already captured by Brent crude oil and the autoregressive structure. Therefore, H3a is rejected and the harvest dummy should not be included in the final predictive model unless later robustness checks show a specific seasonal benefit.

6.6 Model 5: ARIMAX with 2026 Geopolitical Shock

H4_A: The 2026 geopolitical shock represents a structural break where the relationship between oil and ethanol intensifies.

The geopolitical conflict dummy variable was designed to test whether the 2025–2026 geopolitical shock led to an additional upward shift in ethanol prices beyond the effect already captured by lagged Brent crude oil prices. It represents an additive effect to test

if ethanol prices were systematically higher during the conflict period after controlling for oil price movements.

```
--- Model 5: ARIMAX with Brent + Conflict ---  
ARIMAX (Brent+Conflict) AIC: -3419.3767  
ARIMAX (Brent+Conflict) Test RMSE: 0.0577  
  
AIC Improvement over Model 2: -1.9999  
RMSE Improvement over Model 2: -0.0002 (-0.34%)  
  
--- Diebold-Mariano Test: Model 2 vs. Model 5 ---  
DM Statistic: -8.4609  
p-value:      0.0000
```

Figure 6.6.1 - Model 5 Metrics

Model 5 produced an AIC of -3419.3767, compared with -3421.3766 for the Brent-only model. Since the latter has the lower AIC, the conflict dummy also adds complexity without improving the model's in-sample fit enough to justify the additional variable.

The out-of-sample forecasting results also show a slight deterioration. The RMSE increased from 0.0575 in Model 2 to 0.0577 in Model 5, indicating a decline in forecast accuracy of approximately 0.34%.

With a p-value of < 0.05 , the Diebold-Mariano test indicates that the difference in forecast errors between Model 2 and Model 5 is statistically significant. However, as Model 5 has a higher RMSE, it should be interpreted as evidence that adding the conflict dummy worsens forecast performance rather than improves it.

H4a is thus not supported under the additive conflict dummy model as these results suggest the geopolitical shock may already be indirectly captured through Brent crude oil prices. Since Brent reflects global oil market stress, adding a separate binary conflict indicator appears to be redundant in this additive specification.

And since the additive dummy specification may not fully capture the non-linear nature of the shock, the following sections further examine whether the geopolitical crisis

operated through a multiplicative intensification of the oil-to-ethanol transmission mechanism.

The results suggest that a simple additive conflict indicator is insufficient to capture the market dynamics associated with the geopolitical shock. One possible explanation is that the crisis did not merely increase ethanol prices by a fixed amount, but instead altered the strength of the relationship between global oil prices and domestic ethanol markets. Consequently, the next stage of the analysis investigates alternative intervention structures through additive and multiplicative scenario simulations, allowing the shock to influence the forecasting framework in a more flexible manner.

6.7 Structural Break Scenario Simulation: Additive vs. Multiplicative Shock

Additional scenario tests were conducted to compare additive and multiplicative structural break assumptions in order to further evaluate the geopolitical shock hypothesis. The baseline Brent-only model, here referred to as the “peacetime rules” model, produced the lowest RMSE (0.0575) and was thus used as the reference case.

```

--- Testing H4: Additive vs. Multiplicative Structural Break ---
Baseline (Peacetime Rules) RMSE: 0.0575

--- Additive Simulations (Flat Premium) ---
Scenario Add-A (+$0.05) RMSE: 0.0947
Scenario Add-B (+$0.10) RMSE: 0.1401
Scenario Add-C (+$0.20) RMSE: 0.2366

--- Multiplicative Simulations (Oil Sensitivity) ---
Scenario Mult-A (1.3x) RMSE: 0.0630
Scenario Mult-B (1.6x) RMSE: 0.0693
Scenario Mult-C (2.0x) RMSE: 0.0784

VERDICT: The mathematical behavior of the 2026 wartime market most closely matched: Baseline Peacetime Rules

```

Figure 6.7.1 - Scenario Simulation Metrics

The additive scenarios tested whether the conflict period created a fixed risk premium in ethanol prices. In practical terms, this means assuming that the conflict added the same extra amount to ethanol prices every week, regardless of how Brent crude oil moved. Three flat premium scenarios were evaluated: +US\$0.05, +US\$0.10, and +US\$0.20 per liter, representing fixed crisis premiums of 5, 10, and 20 U.S. cents per liter. If this



assumption were correct, adding one of these fixed premiums should have made the forecast closer to the real observed prices during the shock period. All three additive simulations worsened forecast accuracy, increasing RMSE to 0.0947, 0.1401, and 0.2366, respectively. This means that the larger the fixed premium added to the model, the further the forecast moved away from the actual ethanol prices. This suggests that the geopolitical shock did not behave like a simple fixed upward price adjustment.

The multiplicative scenarios tested whether the conflict intensified ethanol's sensitivity to Brent crude oil prices. Instead of assuming that the conflict added a fixed amount to ethanol prices, this approach tests whether ethanol became more responsive to Brent movements during the conflict period. These scenarios increased oil-price sensitivity by 1.3x, 1.6x, and 2.0x, producing RMSE values of 0.0630, 0.0693, and 0.0784, respectively. Although these results were closer to the baseline than the additive simulations, they still performed worse than the Brent-only model when evaluated across the full testing period. However, the fact that the multiplicative scenarios performed much better than the additive scenarios suggests that, if the conflict changed the market relationship, it was more likely by amplifying the Brent-to-ethanol transmission rather than by adding a fixed price premium.

It seems that the geopolitical risk was already reflected in Brent crude oil prices, so, adding a separate conflict premium or manually increasing oil sensitivity introduced unnecessary distortion, as indicated by the scenario analysis, where the Brent-only model remained the strongest forecasting specification. H4a is not supported as a separate full-period forecasting adjustment, even though the results still show that Brent crude oil served as the main channel through which geopolitical risk was transmitted into ethanol price forecasts.

6.8 Isolated Wartime Forecast

Because the full test set includes both normal weeks and conflict weeks, the overall RMSE may not fully show how the model behaves during the actual shock period.

For this reason, this project also looked only at the active conflict weeks. This allowed comparing how each model performed specifically under geopolitical stress, instead of mixing crisis weeks with easier-to-predict normal weeks.

```
--- ISOLATED WARTIME RMSE ---  
Baseline (Peace Rules): 0.0473  
Multiplicative (2.0x): 0.0353  
Additive (+$0.20): 0.1671
```

Figure 6.8.1 - Isolated Wartime Metrics

The results show a different pattern from the full test-set analysis. The baseline Brent-only model had a wartime RMSE of 0.0473, while the multiplicative scenario, where oil sensitivity was increased by 2.0x, achieved a lower RMSE of 0.0353. On the other hand, the additive scenario, which simply added a flat premium of R\$0.20, performed much worse, with an RMSE of 0.1671.

During the conflict period, the shock did not behave like a fixed extra amount added to ethanol prices. Instead, the results indicate that ethanol prices became more sensitive to Brent crude oil movements. In simpler terms, when the market was under geopolitical stress, changes in Brent had a stronger effect on domestic ethanol prices than they did during normal periods.

Therefore, H4a is contextually supported during the isolated wartime period, but the result should be treated as scenario-based evidence rather than definitive statistical validation. The Brent-only model is still the best option for the full testing period as most weeks still follow normal market conditions. However, when isolating only the conflict weeks, the multiplicative scenario performs better. The 2025-2026 geopolitical shock apparently created a context-specific structural break, in which the oil-to-ethanol transmission mechanism strengthened during the crisis.

In summary, H4a is rejected as a general forecasting specification but supported as a context-specific finding: the structural break only manifests when analysis is restricted to

conflict weeks, confirming a regime-dependent intensification of the Brent–ethanol transmission.

6.9 Grid Search and Time-Series Cross-Validation (CV)

While the initial ARIMAX(3,0,0) specification was derived from traditional time-series diagnostics, an additional validation stage was conducted to evaluate whether alternative ARIMA structures could improve forecasting performance.

A systematic grid search was performed on the train data across 50 combinations of autoregressive (p , ranging from 0 to 4), differencing (d , between 0 and 1), and moving-average (q , also ranging from 0 to 4) parameters. Candidate models were evaluated using expanding-window Time Series Cross-Validation (*TimeSeriesSplit*), which preserves the chronological ordering of observations and avoids look-ahead bias.

For each parameter combination, the model was repeatedly estimated on an expanding training window and evaluated on the subsequent validation period. The average Root Mean Squared Error (RMSE), standard deviation of RMSE, AIC, and BIC were recorded to assess model performance and stability - illustrated on Table 6.9.1.

Table 6.9.1 - Top 5 Model Configuration

Rank	(p, d, q)	Average CV RMSE	Std CV RMSE	Average AIC	Average BIC
1	(0, 0, 0)	0.091960	0.023593	-785.456106	-763.279511
2	(0, 0, 1)	0.093469	0.023937	-1131.579682	-1105.706987
3	(0, 0, 2)	0.094810	0.022288	-1345.977681	-1316.408887
4	(0, 1, 0)	0.098400	0.026180	-1617.037513	-1598.576895
5	(0, 0, 3)	0.100788	0.025605	-1471.761087	-1438.496194

Although the cross-validation procedure identified ARIMAX(0,0,0) as the best-performing specification based on average validation RMSE, the model was

subsequently re-estimated using the full training sample and evaluated on the independent holdout dataset (the 20% test set). This final validation step was used to assess whether the cross-validation results generalized to unseen observations.

Despite the superior cross-validation performance of ARIMAX(0,0,0), the original ARIMAX(3,0,0) specification achieved slightly lower forecasting error on the independent holdout sample (Table 6.9.2). Given the forecasting objective of this study, greater weight was assigned to performance on unseen data, and ARIMAX(3,0,0) was retained as the preferred forecasting framework.

Table 6.9.2 - GridSearch Model vs Baseline

Model)	RMSE
Baseline ARIMAX(3, 0, 0)	0.055620
CV-Selected ARIMAX(0, 0, 0)	0.057839

It's worth mentioning that the cross-validation RMSE values are higher than the final holdout RMSE because the expanding-window procedure evaluates the model across multiple historical forecasting windows, including periods with limited training data. Consequently, cross-validation provides a more conservative estimate of forecasting performance.

6.10 Model Comparison Summary

The Table 6.10.1 presents a consolidated comparison of the five ARIMAX specifications estimated across the hypothesis-testing phase, evaluated on two complementary criteria: AIC for in-sample model efficiency and RMSE for out-of-sample forecast accuracy.

Table 6.10.1 - Model Comparison Summary

Metric	Baseline	Model 2 - H1a	Model 3 - H2a	Model 4 - H3a	Model 5 - H4a
AIC	-3425.2527	-3421.3766	-3442.2246	-3419.6672	-3419.3767
RMSE	0.0588	0.0575	0.1328	0.0577	0.0577

The results reveal a consistent divergence between explanatory power and predictive performance across the model sequence. The baseline ARIMAX(3,0,0) provided a strong forecasting benchmark despite relying exclusively on historical ethanol prices and structural-break controls. While it did not achieve the lowest AIC among the competing specifications, it established the reference level against which the incremental contribution of external variables was assessed (-3425.2527).

Model 2, incorporating lagged Brent crude oil prices, represents the best overall forecasting specification. Although its AIC of -3421.3766 is slightly less negative than the baseline (due to the complexity penalty for adding one exogenous variable), it achieved the lowest test RMSE of 0.0575, a 2.24% improvement over the baseline. The Diebold-Mariano test confirmed this improvement is statistically significant ($p = 0.0000$), establishing Model 2 as the preferred economic specification among the competing hypothesis-driven models.

Model 3 produced the most negative AIC of all specifications at -3442.2246, demonstrating that the BRL/USD exchange rate adds substantial explanatory power when fitting historical data. Nevertheless, it generated the highest RMSE (0.1328), with a deterioration of 131.12% relative to Model 2. This result illustrates what is perhaps the most important methodological distinction in the chapter: a variable can be highly significant in-sample while actively degrading out-of-sample accuracy. This means that the exchange rate captures meaningful historical transmission dynamics but introduces volatility that the model cannot reliably project forward.

Models 4 and 5, in turn, tell a similar story. Both the harvest seasonality dummy and the geopolitical conflict dummy worsened AIC relative to Model 2 and produced marginally



higher RMSE values of 0.0577. In both cases, the variables failed to contribute information beyond what the Brent crude oil variable and the autoregressive structure already captured. For the harvest dummy, this reflects the likelihood that predictable seasonal supply cycles are already embedded in the AR(3) lags. For the conflict dummy, the results suggest that an additive conflict specification did not contribute information beyond that already captured by Brent crude oil prices. This finding motivated the subsequent scenario-based intervention analysis presented in Section 6.11.

Taken together, the model comparison supports a clear and economically coherent conclusion: the optimal forecasting specification is one that combines the market's short-term autoregressive memory with the external signal of lagged global oil prices, while excluding variables whose explanatory value does not translate into predictive stability. Model 2 satisfies this criterion and is carried forward as the basis for the residual diagnostics, robustness checks, and final forecast projections presented in the following sections.

Although alternative ARIMA orders were subsequently examined through grid search and time-series cross-validation, the Brent specification remained the strongest-performing economic model and was therefore carried forward into the remaining validation and forecasting exercises.

6.11 Robustness and Sensitivity

Following model selection and cross-validation, additional robustness and sensitivity analyses were performed to evaluate whether the superiority of the Brent specification remained stable across alternative market regimes and forecasting horizons.

6.11.1 Regime stability analysis

In order to verify whether the Brent model was not only performing well due to the 2025-2026 geopolitical crisis, it was also run through a robustness check using only the calmer pre-conflict period. Moreover, removing the conflict window and recalculating the 80/20 train-test split, the Brent-only model continued to exceed the baseline model,



which produced an RMSE of 0.0571, while the Brent model reduced the value to 0.0564, representing a 1.30% reduction in forecasting error.

Brent crude oil seems to have predictive value even during calmer market conditions. In other words, oil prices do not matter only during periods of war or extreme geopolitical volatility. The relationship between Brent and Brazilian ethanol prices appears to be present across different market regimes, although the improvement is smaller in stable periods.

During calm market conditions, domestic factors such as distribution logistics, regional demand, and regular supply patterns may behave more predictably. As a result, the baseline autoregressive model can already capture much of the historical movement in ethanol prices. Brent still helps, but its additional contribution is smaller because the baseline model is not facing the same level of external disruption.

The robustness check supports the use of Brent crude oil as the main exogenous predictor in the forecasting framework. As the RMSE of 0.0564 provides a useful benchmark for model performance under normal operating conditions, it indicates that the model can track weekly ethanol prices with a relatively small forecasting error.

6.11.2 Temporal Robustness Check (2-week and 4-week Ahead Forecasting)

Up to this point, all the models have been built to execute a 1-step-ahead (1-week) dynamic forecast. As the fuel market decisions are not always made only for the following week, and the industry usually operates on bi-weekly or monthly planning cycles, the usefulness and performance of the model were tested for 2-week-ahead and 4-week-ahead predictions.

For the 2-week-ahead forecast, the Brent model achieved the lowest RMSE at 0.0198, compared with 0.0213 for the naive benchmark and 0.0570 for the pure AR(3) model. For the 4-week-ahead forecast, the Brent model again had the lowest RMSE at 0.0293, compared with 0.0310 for the naive benchmark and 0.0579 for the pure AR(3) model. So, the Brent-based model continued to perform better than the alternatives across both forecast horizons.

```

--- True (Rolling-Origin) Robustness Check (Multi-Week Horizons) ---

Evaluating a True Rolling-Origin 2-Week Ahead Forecast Horizon:
-> True Rolling Naïve RMSE:      0.0213
-> Model 1 (Pure AR(3) Rolling) RMSE: 0.0570
-> Model 2 (Brent Rolling) RMSE: 0.0198
SUCCESS: Model 2 maintains structural predictive dominance at horizon t+2!

Evaluating a True Rolling-Origin 4-Week Ahead Forecast Horizon:
-> True Rolling Naïve RMSE:      0.0310
-> Model 1 (Pure AR(3) Rolling) RMSE: 0.0559
-> Model 2 (Brent Rolling) RMSE: 0.0293
SUCCESS: Model 2 maintains structural predictive dominance at horizon t+4!

```

Figure 6.11.2.1 - Temporal Robustness Check Metrics

This suggests that Brent crude oil does not only help with immediate next-week forecasting. It also provides useful predictive information for short-term planning up to four weeks ahead. This is relevant for the project because it shows that the model can support more practical decision-making, not only very short-term prediction.

Even with the Brent model's increased results, expected as the uncertainty over time, the model remained the strongest option among the models tested, reinforcing its role as the main exogenous predictor in the forecasting framework.

6.12 Residual Diagnostic Metrics - Model 2 Brent

After selecting the Brent-based ARIMAX model as the preferred forecasting specification based primarily on its out-of-sample forecasting performance, additional residual diagnostic tests were conducted to assess model adequacy and identify any remaining specification issues.

As illustrated in Figure 6.12.1, the model captures the general behavior of ethanol prices, but it can still struggle with sudden shocks or unusual market conditions. The over time residuals are mostly centred around zero, which suggests that it is not consistently over- or underestimating ethanol prices. During moments of stronger market movement, there are still some larger errors in specific periods.

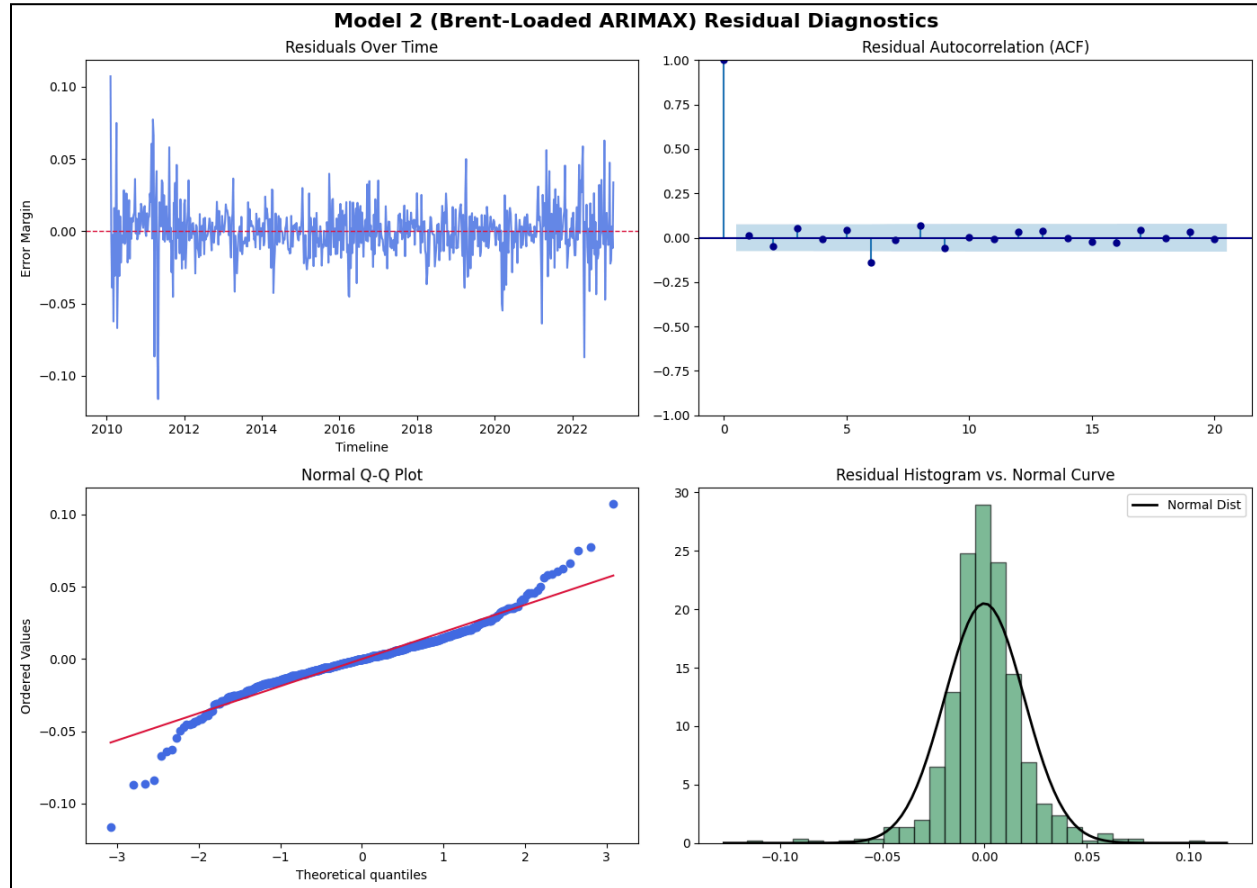


Figure 6.12.1 - Residual Diagnostic Plots

```

--- Ljung-Box Test Metrics (Lag 10) ---
Ljung-Box Chi-Square Statistic: 23.9540
Ljung-Box Test p-value:          0.0077
WARNING: p <= 0.05. Significant autocorrelation remains in the residuals.
        The model may be missing additional lag structures.

```

Figure 6.12.2 - Ljung-Box Test Metrics

The residual autocorrelation plot suggests that most autocorrelation values remain within the confidence bands. However, the Ljung-Box test provides a more rigorous statistical assessment. With a p-value of 0.0077, the null hypothesis of independently



distributed residuals is rejected at the 5% significance level, indicating that some residual autocorrelation remains. This suggests that the model may not fully capture all temporal dependencies present in the ethanol price series.

The Q-Q plot and the histogram show a more mixed result. The majority of residuals are close to the normal distribution around the centre, and the histogram shows that most errors are small and concentrated near zero. It is important to note that the extreme points deviate from the normal line, suggesting that the residuals have heavier tails. This means the model performs better under normal market conditions, but uncertainty increases during extreme events.

Overall, the diagnostic results support the use of the Brent-based ARIMAX model as the primary forecasting framework. The residuals remain centred around zero and exhibit an approximately normal distribution for most observations, indicating that the model successfully captures the main structure of ethanol price dynamics. Nevertheless, the Ljung-Box test reveals residual autocorrelation, while the heavier tails observed in the Q-Q plot suggest reduced accuracy during periods of extreme market volatility. Consequently, the model should be viewed as a strong but not fully exhaustive representation of ethanol price behaviour.

These findings are consistent with the model-selection results presented earlier, where the Brent specification demonstrated superior forecasting performance despite some remaining statistical imperfections.

6.13 Final Forecast Projection

The final graph - Figure 6.13.1 - presents four-week prospective projections of ethanol prices under two different market scenarios. The black line represents the observed historical price of ethanol, while the projected period compares a normal de-escalation scenario with a prolonged crisis scenario.

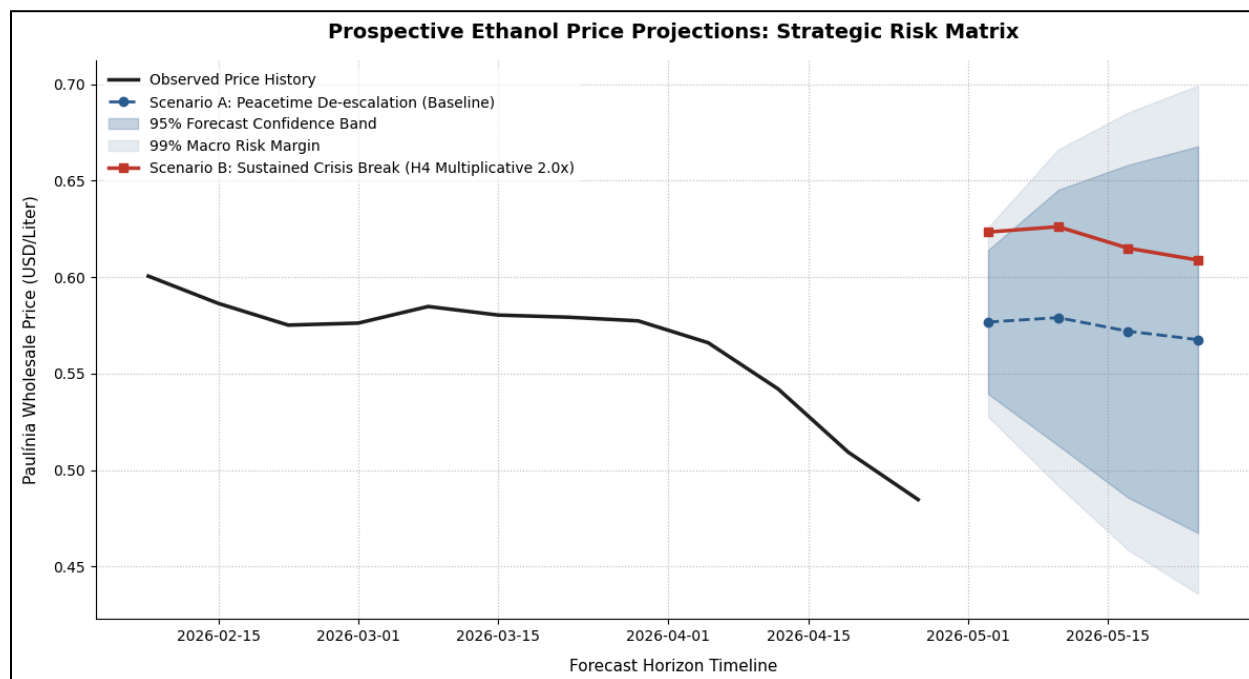


Figure 6.13.1 - Prospective Ethanol Price Projections

Assuming the market continues to follow its normal transmission pattern, where Brent crude remains the main external predictor and without further amplification of the crisis, scenario A considers the peacetime de-escalation baseline, and the Brent-based ARIMAX model projects that ethanol prices will remain around \$0.57–\$0.58 USD/liter, with a slight downward trend over the forecast horizon.

In Scenario B, the sustained-crisis scenario, the multiplicative shock structure identified during the isolated wartime analysis is applied to the baseline forecast. This scenario does not represent a direct statistical prediction of future conflict conditions; rather, it serves as a stress-testing framework for evaluating how continued geopolitical disruptions could alter ethanol price trajectories. Under this scenario, ethanol prices remain higher, around 0.61–0.63 USD/liter at the beginning of the forecast period. This suggests that if geopolitical stress continues, ethanol prices may remain above the baseline forecast because the domestic market becomes more sensitive to Brent crude oil movements.



As expected, forecast uncertainty increases with the prediction horizon. The widening confidence intervals reflect the accumulation of uncertainty as projections extend further into the future. The 95% confidence band represents the expected forecast range under the model's normal uncertainty assumption, while the wider 99% macroeconomic risk margin reflects the possibility of larger deviations under unstable market conditions.

Overall, the forecasting framework demonstrates value beyond point prediction. By combining a statistically validated baseline forecast with scenario-based intervention analysis, the model provides decision-makers with a range of plausible outcomes under alternative market conditions. The baseline scenario reflects expected behavior under normal or de-escalating conditions, whereas the sustained-crisis scenario illustrates the potential consequences of prolonged geopolitical stress. This dual-perspective approach supports more informed planning by highlighting both expected market behavior and downside risk exposure.

7. Limitations

While the developed forecasting framework presents a somewhat robust out-of-sample performance and confirms the transmission of global crude oil shocks to local ethanol prices, a rigorous econometric deployment requires a transparent evaluation of its boundary conditions. Modeling complex, regulated commodity markets inherently involves trade-offs between mathematical parsimony and real-world systemic shifts.

The limitations outlined below do not invalidate the predictive utility of the selected model, but they address its operational envelope. By examining geographic abstractions, historical geopolitical anomalies, residual issues, and the divergence between explanatory and predictive variables, this section seeks to address the necessary caution for interpreting the model's outputs and guiding future enhancements.

Limited Naive Benchmarking:

Although naive benchmarks were incorporated during the temporal robustness analysis, they were not included in the primary one-step-ahead model selection framework. Future research could extend the comparison by systematically benchmarking all candidate specifications against random-walk and seasonal-naive forecasting models throughout the entire model-building process.

Geographic Concentration of Price Data:

The use of Paulínia wholesale prices as the ethanol price indicator reflects data availability and market centrality, but may not fully capture price dynamics in Brazil's North and Northeast regions, where ethanol supply chains are thinner, transport costs are higher, and the ethanol-to-gasoline ratio frequently differs materially from the national average.

2026 Geopolitical Oil Shock as a Confounding Factor:

The Iran–US conflict introduced a supply shock of unprecedented magnitude in the final months of the dataset. Models trained predominantly in the 2010 – 2025 period may exhibit reduced out-of-sample validity when applied to post-shock conditions. This



limitation was partially addressed through sensitivity analyses, intervention-based scenario simulations, and isolated wartime forecasting exercises. Nevertheless, any future geopolitical disruption may differ substantially from the conditions observed during the study period.

Residual Autocorrelation and Model Specification:

Although the Brent-based model performed well compared with the other models, the residual diagnostics show that it is not perfectly specified. The Ljung-Box test produced a p-value below 0.05, meaning that some autocorrelation still remains in the residuals.

The model may be missing additional lag effects, nonlinear relationships, and explanatory variables not included in the current framework. Residual autocorrelation suggests that time-related structure may persist in the errors, indicating that the model is useful for forecasting; the residuals are not pure white noise, and the results should be interpreted with some caution.

Extreme Errors and Heavy Tails:

Fuel markets can strongly react to geopolitical shocks, inflationary pressure and sudden commodity price movements. Under normal conditions, the model tracks ethanol prices reasonably well. However, during periods of extreme volatility, the forecast uncertainty becomes higher, and the model can underestimate the size of possible price movements. As shown in the Q-Q plot and histogram, errors are small and concentrated around zero, but the extreme values move away from the normal distribution, suggesting the residuals have heavier tails and the model can produce larger errors during unusual periods.

Exchange Rate as Exploratory:

As Brent crude oil is priced in US dollars, changes in the exchange rate can affect how those international price movements are translated into domestic prices. Despite the significant difference between explanation and prediction, the exchange rate improved the in-sample fit, based on AIC, helping explain historical data. It also worsened the out-of-sample RMSE, explaining the non-improvement in future forecasting variables.



Not every economically relevant variable improves prediction. Some variables help explain the past, but they can still make the model less reliable when forecasting unseen data.

Cross-Validation and Holdout Divergence:

The expanding-window cross-validation procedure identified ARIMAX(0,0,0) as the best-performing specification based on average validation RMSE. However, when evaluated on the independent holdout sample, the original ARIMAX(3,0,0) model achieved slightly superior forecasting performance. This divergence illustrates that model-selection procedures may not always generalize perfectly to future observations, particularly when structural changes occur near the end of the sample period. Consequently, forecasting performance should be interpreted using multiple validation approaches rather than relying exclusively on a single optimization criterion.

8. Business Impact

The forecasting framework developed in this project is designed not only to advance knowledge of ethanol price dynamics but to generate outputs that are immediately actionable for the stakeholders who manage Brazil's dual-fuel energy economy. The Brent-based ARIMAX model produces a weekly price direction signal that is expected to be embedded into four distinct decision-making contexts: supply chain optimization for fuel distributors, retail pricing strategy for gas station operators, inflation monitoring for the Central Bank, and subsidy budget planning for the Ministry of Finance. Each application draws from a different aspect of the model's architecture, such as its 1-week-ahead accuracy, its 2- to 4-week temporal robustness, and its scenario-based risk matrix, connecting the statistical outputs directly to real operational choices.

The practical value of the model extends beyond a single forecast number. Because the framework produces both a point estimate and confidence intervals under two market scenarios (a peacetime de-escalation baseline and a sustained geopolitical stress scenario), decision-makers can align their response to whichever risk regime is most plausible at any given time. During normal market conditions, the baseline forecast is sufficient, whereas during periods of geopolitical stress or oil market volatility, the multiplicative crisis scenario provides a more conservative planning boundary. The sections below describe how each stakeholder can operationalise these outputs.

8.1 Supply chain optimization: Petrobras and regional distributors

For Petrobras (Brazil's national oil company) and regional fuel distributors, the model can support short-term ethanol supply planning by providing a weekly price direction signal. Although the fuel market does not operate only through weekly decisions, the 1-week-ahead forecast can still help supply managers anticipate whether prices are expected to rise, fall, or remain stable in the near term.

This predictive insight allows Petrobras to optimize its logistics before consumers and retailers react to shifting market dynamics. For instance, if the model projects a price hike, Petrobras can ramp up deliveries early, keeping distribution hubs well-stocked

before the price increase fully ripples through the market. Conversely, ahead of a projected low-price week, managers can strategically delay non-urgent shipments or hold off on building up expensive, unnecessary inventory.

8.2 Retail pricing strategy: gas station operators

For independent gas station owners in Brazil's flex-fuel market, the primary challenge is managing profit margins rather than logistics. Their success depends on the spread between wholesale purchase costs and pump prices, especially since consumers constantly choose between ethanol and gasoline based on which is cheaper per kilometer. Because the ethanol-to-gasoline price ratio determines what drivers buy, weekly wholesale price shifts can make or break retail demand.

The 1-week-ahead forecast gives retailers a crucial head start to time their wholesale purchases:

- **When a price increase is forecasted:** retailers get a window to buy extra stock at current lower rates. This short-term buffer lets them keep their pump prices competitive and protects their margins while unprepared competitors are forced to hike prices immediately.
- **When a price drop is forecasted:** retailers minimize their inventory, delay large wholesale orders until prices fall, and accept low stock levels rather than holding expensive fuel that is losing value. The model's 95% confidence intervals help them gauge how certain the drop is so they can size their inventory cuts accordingly.

The model also serves as an early warning system for fuel parity crossings. When ethanol prices approach the critical 70% threshold of gasoline cost, consumer switching behavior shifts rapidly. If the forecast shows prices nearing this boundary, retailers can prep their pricing and marketing strategies ahead of time.

However, because the model is anchored to the Paulínia wholesale benchmark, its direct numbers are less accurate for gas stations in Brazil's North and Northeast regions. Due to higher freight costs and thinner local supply chains in those areas,

retailers there need to apply a regional price adjustment rather than using the raw forecast out of the box.

8.3 Inflation forecasting: Central Bank of Brazil

Fuel prices are incredibly visible in Brazil and heavily impact the country's IPCA consumer price index. Because ethanol and gasoline costs directly affect travel and indirectly drive up food prices through distribution, any sustained change in wholesale ethanol prices ripples into broader inflation within one to four weeks. For the Central Bank of Brazil, having an early-warning system for these fuel shocks is quite helpful for shaping monetary policy and guiding interest rate decisions.

The model can help policymakers by turning weekly ethanol forecasts into monthly rolling averages that act as a fuel-price sub-index. If the model predicts three or more consecutive weeks of rising prices, the Central Bank can flag it as a lasting structural threat to inflation rather than just temporary market noise. The model's dual-scenario setup makes it easy to tell the difference between a quick spike that will fix itself in a couple of weeks and a persistent problem that requires a real policy response.

In practice, the inflation desk can use these scenarios directly:

- **Scenario A (Peacetime):** if the baseline shows stable or falling ethanol prices, economists can keep their current inflation forecasts steady.
- **Scenario B (Sustained Crisis):** if prices stay elevated, which aligns with findings that domestic market sensitivity to Brent crude doubles during geopolitical crises, they can proactively raise its inflation assumptions. This allows them to discuss policy rate adjustments before the price hikes completely spread through the economy.

This foresight is crucial, especially since the World Bank's April 2026 Commodity Markets Outlook warned that developing economies could see inflation hit 5.8% in 2026 under prolonged disruption scenarios. The model helps the Central Bank see whether a global oil spike is being safely absorbed by local ethanol buffers, as seen during the 2026 conflict when Brazilian fuel prices only rose 10% to 20% compared to 30% to 40%

in the U.S., or if prices are breaking through those buffers. Catching this early prevents reactive, late interventions that happen after inflation expectations have already spun out of control

8.4 Subsidy Budget Planning: Ministry of Finance

For the Ministry of Finance, the model can support fuel subsidy planning and fiscal decision-making. During periods of fuel price pressure, governments may face pressure to introduce subsidies, tax reductions, or other consumer protection measures. However, these policies can be expensive if used unnecessarily.

The model can help differentiate between temporary price spikes and more persistent structural increases. If prices are expected to rise for only a short period and then normalise, the government may decide not to activate a subsidy and instead preserve fiscal space. If it indicates a sustained increase through several weeks, the Ministry can begin preparing a targeted fiscal response earlier.

This is one of the most important policy applications of the model. It allows the government to move from reactive crisis management to planned intervention. Instead of responding only after prices have already increased, policymakers can use the forecast to evaluate whether intervention is necessary, when it should be activated, and how large the fiscal response may need to be.

8.5 Stakeholder Decision Framework - Summary

Table 8.5.1 - Stakeholder Decision Framework

Stakeholder	Decision Context	Forecast Application	Planning Horizon	Expected Operational Outcome
Petrobras - Supply Chain	Weekly ethanol delivery volumes and	Use 1-week-ahead forecasts to anticipate	1-week lead time	Reduced holding costs, optimised truck routing, and supply matched to

	distribution routing	price direction; adjust delivery schedules and inventory allocation proactively		demand before market reacts
Retail Gas Station Owners	Weekly pump pricing and stock purchase decisions	Use wholesale price forecasts to plan purchase timing, set competitive pump margins, and design promotional windows	1-week lead time	Lower inventory risk, improved margin protection, and better promotional timing relative to competitors
Central Bank of Brazil	Monthly inflation forecast inputs and interest rate guidance	Aggregate weekly ethanol forecasts into monthly fuel price indicators; flag persistent	1–4 week rolling horizon	Earlier detection of fuel-driven inflationary pressure; more forward-looking monetary policy deliberation

		vs. transitory price spikes for IPCA modelling		
Ministry of Finance	Fuel subsidy budget allocation and fiscal contingency planning	Assess whether a price elevation is structural or temporary; calibrate subsidy trigger thresholds accordingly	1–4 week rolling horizon	Avoidance of premature or unnecessary subsidy activation; fiscal space preserved for sustained shocks requiring targeted intervention

9. Conclusion

This project set out to determine which macroeconomic factors (among Brent crude oil prices, the BRL/USD exchange rate, harvest seasonality, and a geopolitical shock indicator) contribute most to explaining the weekly variation in Brazilian wholesale ethanol prices. The incremental ARIMAX framework provided strong evidence that Brent crude oil is the most useful forecasting variable among those evaluated. Operating through a robust one-week transmission lag, this relationship remains consistent across market regimes evaluated in this study, holds true when excluding crisis periods, and maintains predictive stability over multi-week forecast horizons.

Additional validation through expanding-window cross-validation and an independent holdout test reinforced the robustness of the forecasting framework, with the original ARIMAX(3,0,0) specification maintaining superior performance on unseen observations despite alternative candidate structures identified during optimization.

The evaluation of alternative hypotheses yielded critical insights regarding the behavior of explanatory variables in a time-series model. The BRL/USD exchange rate demonstrated that strong in-sample explanatory power during training can substantially degrade out-of-sample forecasting accuracy. Due to temporary currency decoupling during crisis periods, the exchange rate introduces volatility that cannot be reliably projected forward, confirming it is better utilized as a structural interpretation tool rather than a raw forecasting input. Meanwhile, harvest seasonality proved redundant, as the predictable annual supply cycle of sugarcane is already efficiently absorbed by the model's internal autoregressive lag memory. Finally, the additive geopolitical conflict indicator encountered a zero-variance training condition because the shock occurred entirely within the out-of-sample window, proving that purely statistical frameworks remain blind to unprecedented structural breaks unless augmented by intentional scenario analysis.

Translating this structural blind spot into actionable insight, the scenario analysis and isolated wartime evaluations demonstrated how geopolitical risk actually maps to the domestic energy market. While the peacetime Brent model functions best as a



general-purpose specification across the full time series, the 2026 geopolitical shock did not add a simple flat price premium. Instead, it functioned as a multiplier that dramatically intensified the domestic market's sensitivity to Brent crude movements, transmitting geopolitical risk directly through the global oil channel. Robustness checks further validated this dynamic, showing that the model retains its predictive advantage in calm periods and holds multi-week validity at both 2-week and 4-week horizons, elevating the framework from an academic exercise to a deployable planning tool.

From a practical perspective, this framework generates immediate, actionable value for key stakeholder groups managing Brazil's energy, inflation, and fiscal policies. For Petrobras and regional distributors, the model provides a directional signal to optimize supply logistics and inventory allocation; for gas station retailers, it enables smarter purchase timing and margin protection. On a macroeconomic level, the Central Bank can aggregate these weekly forecasts into monthly indices for proactive inflation monitoring, while the Ministry of Finance can leverage the four-week horizon and multiplicative shock insights to design and time responsive fuel subsidy mechanisms.

While the model functions effectively, residual diagnostics, such as remaining autocorrelation and heavy distribution tails during crisis events, demand caution when interpreting confidence intervals during periods of extreme market stress. Although residual diagnostics revealed some remaining autocorrelation and heavier tails during crisis periods, these limitations did not overturn the central finding regarding the predictive importance of Brent crude oil prices. Furthermore, limitations such as a geographic focus on Paulínia and an inability to automatically learn unprecedented shocks present clear pathways for future research.

Future research could explore the inclusion of localized variables such as regional freight costs, taxation policies, refinery margins, and inventory indicators. Additional work may also compare the ARIMAX framework against alternative forecasting approaches, including machine-learning architectures such as Gradient Boosting, Random Forests, and Long Short-Term Memory (LSTM) networks. Ultimately, the Brent-based ARIMAX framework fulfills the objectives of this study by providing a



practical, economically interpretable, and operationally deployable forecasting tool. While it cannot eliminate uncertainty or anticipate unprecedented shocks automatically, it offers a structured analytical framework capable of supporting more informed decision-making in Brazil's fuel and energy markets.

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